

Expected Questions Youll Be Able to Answer (Learning Outcomes)

1. Binary Basics

- What is a bit and a byte?
- How do computers represent data using only 0 and 1?
- How is a number like 13 stored in binary?

2. Encoding & Characters

- What is ASCII, and how does it represent characters like A, z, or @ in binary?
- What is the difference between ASCII, UTF-8, and Base64?
- How is a string like "hello" stored in memory?

3. Data Conversion

- How do you convert:
 - A binary number to decimal?
 - A decimal number to binary?
 - A string to binary using Node.js?

4. Node.js and Binary

- What is a Buffer in Node.js?
- How can you read/write binary data from/to a file using fs.read() and Buffer?
- How does Buffer.from("hello") store that string in memory?

5. Hex & Byte-level Understanding

- What is a hexadecimal number and why is it used?
- How do you convert binary to hexadecimal and vice versa?
- How does Buffer.toString("hex") work?

6. Bitwise Operations

- What does a & b or a | b do in binary?
- How can you use bitwise operations to:
 - Toggle a bit?
 - Set/clear a flag?
 - Check permissions?

7. Endianness

- What is Endianness (Little Endian vs Big Endian)?
- Why does `buffer.readUInt32LE()` behave differently than `readUInt32BE()`?
- When do you need to care about byte order in Node.js?

8. Real-world Representation

- How is an image or audio file stored in binary?
- How is a JSON file stored differently from a PNG file?
- Why do binary files break when opened in a text editor?

Bonus Outcomes (Advanced Dev Perspective)

- Why does encoding mismatch cause "" characters?
- Why can `Buffer.length` be different from `string.length`?
- Why are bitwise operations preferred in low-level protocols?